

The Rise of Mobile Application Development Platforms

There is a huge market for enterprise-ready mobile cross-platform development tools as a result of the increasing diversity in the mobile device ecosystem. Enterprises of all sizes are craving for solutions that allow the development of scalable applications using a single code base, while also taking full advantage of the capabilities of the device. The market is large, but developers few!

The use of mobile computers is spreading faster than any other consumer technology in history. Smartphones outsell PCs. Touchscreens outnumber keyboards.

As per the Strategy Analytics Research Report on 'Global Mobile Workforce Forecast Update 2012-2018', worldwide, the mobile workforce is predicted to increase from 1.26 billion in 2013, accounting for 36.4 per cent of the global workforce, to 1.67 billion in 2018, accounting for 41 per cent of the global workforce.

Clearly, mobile apps are now being viewed by enterprises as core drivers that improve operational efficiency and have qualified as business-critical technology. However, app delivery is less about the technical aspects of programming and more about an app's business impact, design, integration, deployment and management.

This article will delve into the major challenges involved in designing, developing, deploying and maintaining mobile apps for enterprises, and how mobile application development platforms (MADP) solve some of these challenges.

Enterprise mobile apps: The challenges

Enterprise mobile apps involve some specific challenges that need to be addressed right in the planning and design phases. Later discoveries of one or more of these challenges have resulted in many enterprises having to re-plan and redesign their mobility solution roadmap. Let's take a deeper look at these challenges.

BYOD and device diversity

The Bring Your Own Device (BYOD) trend has penetrated

enterprises everywhere. If your enterprise mobile app is to support the BYOD strategy, then an altogether different level of challenges needs to be addressed. The target device spectrum grows exponentially when considering BYOD, with new devices entering the market and reaching your target users every day. Keeping track of mobile platforms, versions and new device models (on a variety of platform versions and differing configurations) makes supporting BYOD for your enterprise app a big challenge, without a well-planned approach.

Even if you do not opt for BYOD, the target device spectrum may be limited yet heterogeneous. Diverse mobile platforms (Android, iOS, etc), along with the dynamics they bring in, makes app development for supporting multiple platforms complex. Further, apart from the variety of devices that currently exist, more keep arriving in the market for each of the mobile platforms, thereby imposing the need for an enterprise to ensure that the mobile apps not only support required mobile platforms, but also work well on the target device spectrum per platform.

Enterprise data access

An enterprise mobile app may need to access data stored in a variety of heterogeneous data sources at the enterprise backend. For instance, the sales team's data may reside in the Salesforce CRM while the financial data resides in Oracle. Integrating with a variety of standard and custom data sources may be a big need when creating an enterprise mobile app.

